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[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

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MCA-6105 (Python Programming)

MCA-1st NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

Section A

1. Explain the basic data types in Python with suitable examples. How are conditional statements and looping statements implemented in Python? (12)
2. Write a Python program that accepts a list of integers and performs the following:
 - 1) Finds the maximum and minimum number
 - 2) Reverses the list
 - 3) Prints only even numbers (3×4=12)

Section B

3. Explain the concepts of inheritance, polymorphism, and operator overloading in Python with examples. (12)
4. Write a program in Python to read a configuration file, modify its content, and write the updated data into a new file. (12)

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Section C

5. Discuss exception handling in Python. Write a program to handle division by zero and file-not-found errors. (12)
6. Explain how database operations are performed in Python. Write a program to connect to an SQL database, create a table, and insert at least three records. (12)

Section D

7. How can Python be used for sending bulk emails in marketing? Write a Python program using the **smtplib** module to send a mail to multiple users. (12)
8. Write short notes with examples on the following Python libraries:
 - 1) SciPy
 - 2) NumPy
 - 3) Matplotlib (3×4=12)

Section-E (Compulsory)

9. Short Answer Questions (6×2=12)
 - (a) List two advantages of Python over other programming languages.
 - (b) What is a destructor in Python?
 - (c) Differentiate between `write()` and `writelines()` functions.
 - (d) Mention any two benefits of exception handling in Python.
 - (e) Name two GUI frameworks available in Python.
 - (f) Give one real-life application each of Matplotlib and SciPy. (6×2=12)